MongoDB Compass - Local Mongo/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

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mongosh: Local Mongo | mongosh: Local Mongo | mongosh: Local Mongo | mongosh: Local Mongo

>_MONGOSH

```
> use collegeDB
< switched to db collegeDB
> show collections
< activities
  courses
  enrollments
  faculty
  students
> db.students.find(
    {
      attendance: { $gt: 85 },
      skills: { $all: ["MongoDB", "Python"] }
    },
    {
      _id: 0,
      name: 1,
      department: 1
    }
  )
<
collegeDB> //The output was empty because no student had both "MongoDB" and "Python" skills present together in the skills array.
```

```
>_MONGOSH
> db.faculty.aggregate([
    {
      $project: {
        name: 1,
        totalCourses: { $size: "$courses" }
      }
    },
    {
      $match: {
        totalCourses: { $gt: 2 }
      }
    },
    {
      $project: {
        _id: 0,
        name: 1,
        totalCourses: 1
      }
    }
  ])
< {
    name: 'Charles Newton',
    totalCourses: 3
  }
  {
    name: 'Julia Cole',
    totalCourses: 3
  }
  {
```

```
>_MONGOSH
> db.enrollments.aggregate([{ $lookup: { from: "students", localField: "student_id", foreignField: "_id", as: "student" } }, { $unwind: "$student" },
< {
    courses: [
      'Customizable client-driven secured line',
      'Focused client-server knowledge user'
    ],
    name: 'Vincent Norris'
  }
  {
    courses: [
      'Sharable bifurcated paradigm',
      'Digitized even-keeled Internet solution',
      'Proactive optimizing initiative',
      'Customizable client-driven secured line'
    ],
    name: 'Megan Taylor'
  }
  {
    courses: [
      'Configurable global framework',
      'Multi-lateral systemic success'
    ],
    name: 'Janet Jarvis'
  }
  {
    courses: [
      'Organic incremental task-force'
    ],
```

```
>_MONGOSH
> db.enrollments.aggregate([{ $lookup: { from: "courses", localField: "course_id", foreignField: "_id", as: "course" } }, { $unwind: "$course" }, { $
< {
    totalStudents: 2,
    avgMarks: 94.5,
    courseTitle: 'Total tangible moderator'
  }
  {
    totalStudents: 1,
    avgMarks: 82,
    courseTitle: 'Open-architected tangible protocol'
  }
  {
    totalStudents: 1,
    avgMarks: 86,
    courseTitle: 'Advanced analyzing budgetary management'
  }
  {
    totalStudents: 1,
    avgMarks: 71,
    courseTitle: 'Realigned scalable extranet'
  }
  {
    totalStudents: 1,
    avgMarks: 92,
    courseTitle: 'Quality-focused local leverage'
  }
  {
    totalStudents: 1,
```

```
>_MONGOSH
    totalStudents: 1,
    avgMarks: 51,
    courseTitle: 'Enhanced radical secured line'
  }
  {
    totalStudents: 1,
    avgMarks: 78,
    courseTitle: 'Balanced asynchronous framework'
  }
  {
    totalStudents: 1,
    avgMarks: 95,
    courseTitle: 'Seamless high-level installation'
  }
  Type "it" for more
> db.enrollments.aggregate([{ $group: { _id: "$student_id", avgMarks: { $avg: "$marks" } } }, { $sort: { avgMarks: -1 } }, { $limit: 3 }, { $lookup:
< {
    avgMarks: 100,
    name: 'Diane Phillips'
  }
  {
    avgMarks: 98,
    name: 'Brandon Rios'
  }
  {
    avgMarks: 94,
    name: 'Christopher Benson'
  }
collegeDB >
```

```
>_MONGOSH
    avgMarks: 78,
    courseTitle: 'Balanced asynchronous framework'
  }
  {
    totalStudents: 1,
    avgMarks: 95,
    courseTitle: 'Seamless high-level installation'
  }
  Type "it" for more
> db.enrollments.aggregate([{ $group: { _id: "$student_id", avgMarks: { $avg: "$marks" } } }, { $sort: { avgMarks: -1 } }, { $limit: 3 }, { $lookup:
< {
    avgMarks: 100,
    name: 'Diane Phillips'
  }
  {
    avgMarks: 98,
    name: 'Brandon Rios'
  }
  {
    avgMarks: 94,
    name: 'Christopher Benson'
  }
> db.students.aggregate([{ $group: { _id: "$department", totalStudents: { $sum: 1 } } }, { $sort: { totalStudents: -1 } }, { $limit: 1 }, { $project:
< {
    totalStudents: 23,
    department: 'Electrical'
  }
collegeDB>
```

```
>_MONGOSH

> db.students.updateMany({_id: {
        $in: db.activities.distinct("student_id", { activity: "Hackathon", result: "Won" })
      }},{
      $set: { attendance: 100 }
    })
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0
  }
> db.activities.deleteMany({year: { $lt: 2022 }})
< {
    acknowledged: true,
    deletedCount: 0
  }
> db.courses.updateOne(
    { _id: "C150" },
    {
      $set: {
        title: "Advanced Data Structures",
        credits: 4
      }
    },
    { upsert: true }
  )
< {
    acknowledged: true,
    insertedId: 'C150',
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 1
  }
collegeDB>
```

```
>_MONGOSH
> db.students.find(
    {
      skills: { $in: ["Python"], $nin: ["C++"] }
    },
    {
      _id: 0,
      name: 1,
      department: 1
    }
  )
< {
    name: 'Kyle Hale',
    department: 'Electrical'
  }
  {
    name: 'Cody Whitehead',
    department: 'Biotechnology'
  }
  {
    name: 'Thomas Jackson',
    department: 'Electrical'
  }
  {
    name: 'Steven Wong',
    department: 'Biotechnology'
  }
  {
    name: 'Cheryl Jackson',
    department: 'Civil'
> db.activities.aggregate([{ $match: { type: { $in: ["Seminar", "Hackathon"] } } }, { $group: { _id: "$student_id", activities: { $addToSet: "$type"
< {
    name: 'Taylor Webb'
  }
  {
    name: 'Patricia Scott'
  }
  {
    name: 'Lydia Day'
  }
  {
    name: 'Adam Solomon'
  }
  {
    name: 'Carlos Bryant'
  }
collegeDB>
```

```
>_MONGOSH
    student_name: 'Michelle Walters',
    department: 'Computer Science',
    course: 'Web Development'
  }
> db.enrollments.aggregate([{
      $lookup: {
        from: "students",
        localField: "student_id",
        foreignField: "_id",
        as: "student"
      }
    },{$unwind: "$student"},{
      $lookup: {from: "courses",
        localField: "course_id",
        foreignField: "_id",
        as: "course"
      }
    },  {$unwind: "$course"},
    { $match: {
        marks: { $gt: 80 },
        "student.department": "Computer Science",
        "course.title": "Web Development"
      }},{$project: {_id: 0, student_name: "$student.name",
        department: "$student.department",
        course: "$course.title",
        marks: 1
      }}])
<
collegeDB>
```

```
> db.activities.aggregate([
    {$group: {
        _id: "$type",
        totalParticipants: { $sum: 1 }
      }
    }, {$sort: { totalParticipants: -1 }},{$limit: 1},
    {$project: {
        _id: 0,
        activity_type: "$_id",
        totalParticipants: 1
      }
    }
  ])
< {
    totalParticipants: 35,
    activity_type: 'Hackathon'
  }
collegeDB>
```